1 **(a)** $\frac{1}{2}(10+16)\times 8$ **[M1]**

104 **[A1]**

(b) Base of right-angled triangle $= 16-10=6$ **[M1]**

Slanted edge $=\sqrt{6^2+8^2}=10$ **[M1]**

$8+10+16+10=44$ **[A1]**

[Total Marks: 5]

2 **(a)** $2(2x+3)+2(x-1)=34$ OR $2x+3+x-1=17$ **[M1]**

$6x+4=34$ OR $3x+2=17$ **[M1]**

$x=5$ **[A1]**

(b) Substitution of $x=5$: $(2(5)+3)\times(5-1)$ OR 13×4 **[M1]**

52 **[A1]**

[Total Marks: 5]

3 **(a)** Area of rectangle $=4(2x+3)$ OR $8x+12$ **[M1]**

Area of triangle $=\frac{1}{2}\times 6\times(3x-1)$ OR $9x-3$ **[M1]**

$8x+12=9x-3$ **[M1]**

$x=15$ **[A1]**

(b) Length $=2(15)+3=33$ **[M1]**

Perimeter $=2(4+33)=74$ **[A1]**

[Total Marks: 6]

4 **(a)** $\frac{1}{2}(x+x+4)(x-1)=40$ **[M1]**

$\frac{1}{2}(2x+4)(x-1)=40$ OR $(x+2)(x-1)=40$ **[M1]**

$x^2+2x-x-2=40 \implies x^2+x-42=0$ **[A1]**

(b) $(x+7)(x-6)=0$ **[M1]**

6 **[A1]**

(c) Identifies lengths are Top $=6$, Bottom $=10$, Height $=5$ **[M1]**

Horizontal difference $=4$, so slanted side $=\sqrt{5^2+4^2}$ OR $\sqrt{41}$ **[M1]**

Perimeter $=5+6+10+\sqrt{41}$ **[M1]**

27.4 (or $27.40\ldots$) **[A1]**

[Total Marks: 9]

5 **(a)** $\frac{1}{2} \times (15 + 24) \times 8$ **[M1]**

156 **[A1]**

(b) Horizontal distance from D to the altitude from $A = \sqrt{10^2 - 8^2} = 6$ **[M1]**

Horizontal distance for the right-angled triangle under $B = 24 - 15 - 6 = 3$ **[M1]**

$BC = \sqrt{8^2 + 3^2}$ $(= \sqrt{73} \approx 8.544)$ **[M1]**

57.5 OR 57.54 to 57.55 **[A1]**

[Total Marks: 6]

6 **(a)** Perimeter of rectangle $= 6y$ OR Perimeter of square $= 60 - 6y$ **[M1]**

$\frac{60 - 6y}{4}$ OR $15 - 1.5y$ **[A1]**

(b) $(15 - 1.5y)^2 + y(2y) = 113$ **[M1]**

$225 - 45y + 2.25y^2 + 2y^2 = 113$ **[M1]**

$4.25y^2 - 45y + 112 = 0$ leading to $17y^2 - 180y + 448 = 0$ with no errors seen **[A1]**

(c) $\frac{-(-180) \pm \sqrt{(-180)^2 - 4(17)(448)}}{2(17)}$ **[M1]**

$\frac{180 \pm \sqrt{1936}}{34}$ OR $\frac{180 \pm 44}{34}$ **[M1]**

$y = 4$ and $y = 6.59$ (or 6.588... or $\frac{112}{17}$) **[A1]**

(d) Identifies $y = 4$ as the valid solution (since $15 - 1.5(6.59) < 6.59$) leading to side length $= 9$ **[M1]**

81 **[A1]**

[Total Marks: 10]